Congestion Games, Player-Specific Congestion Games and Weighted Congestion Games for Radio Resource Management in Wireless Networks

Vaggelis G. Douros, Stavros Toumpis and George C. Polyzos
Mobile Multimedia Laboratory, Department of Informatics
Athens University of Economics and Business
Patision 76, 10434, Athens, Greece
E-mail: {douros, toumpis, polyzos}@aueb.gr

Abstract

The goal of this report is two-fold: Firstly, we present a short introduction on congestion games, a class of non-cooperative games that have received considerable attention by the research community during the last years. We discuss their properties and present their two most prominent extensions, namely weighted congestion games and player-specific congestion games. Secondly, we discuss applications of these games for radio resource management in wireless networks, such as in spectrum sharing in multichannel cellular systems.

I. Introduction and Motivation

Congestion games are a well established approach to model scenarios in which selfish agents individually strive to allocate resources as effectively as possible. They have been introduced by Rosenthal [1] in 1973 and can be described as follows. In a congestion game we are given a finite set of resources and a finite set of players. Each player comes along with a set of strategies, the so-called player's strategy space. The strategies are the available options the player has and each of them corresponds to a subset of the set of resources. For each resource we are also given a payoff (latency) function as specified by the combination of strategies played. Now each player individually has to choose a strategy that maximizes his payoff, (equivalently, minimizes his latency) given the choices of the other players.

Some common assumptions for the formulation of a (congestion) game are the following: Players are considered selfish, *i.e.*, they do not care about each other. Additionally, it is assumed that players have complete knowledge about the game meaning each player knows the entire set of resources, his complete strategy space, his latency functions and in every state of the game the choices of all other players. Furthermore, it is assumed that the players act rationally meaning that their behavior is uniquely determined by their latencies, so that, a player which changes his strategy always switches to a strategy

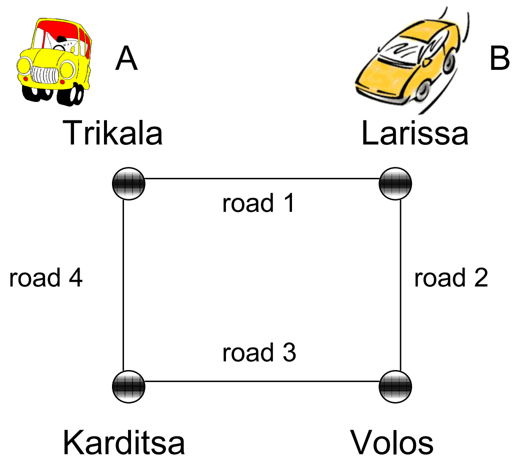

Fig. 1: An example of a congestion game.

strictly decreasing his latency. More specifically, players sequentially switch to the best available strategy given fixed choices of the others. This is called a sequential best response dynamics scheme. Under these assumptions, a Nash Equilibrium (NE) is defined as a state in which none of the players can unilaterally decrease his latency by changing his strategy given fixed choices of the others.

The following example, which is adapted from [2] will help us visualize the notion of congestion games. In Fig.1, there are 2 players (driver A and driver B) that share the following resources (roads): road 1, road 2, road 3, and road 4. Each player should choose a subset of these roads so as to reach his destination. Let's suppose that player A wants to go to Volos and player B to Karditsa. The strategy set S for player A is S_A={{road 1, road 2}, {road 4, road 3}}, whereas the strategy set for player B is S_B={{road 1, road 4}, {road 2, road 3}}. The latency l_i of each resource i is given in Table I and depends on the congestion of the selected resources, which is a function of the number of players that are going to choose them.

TABLE I: Latency of each resource of the game.

$l_1(1) = 2$	$l_1(2) = 3$
$l_2(1) = 1$	$l_2(2) = 4$
$l_3(1) = 2$	$l_3(2) = 3$
$l_4(1) = 2$	$l_4(2) = 5$

Then, if, e.g., player A chooses the strategy "road 1, road 2" and player B chooses the strategy "road 1, road 4" (for the simplification of the symbols, we will omit the word "road"), then, at that particular state:

$$l_A(1,2) = l_1(2) + l_2(1) = 4, \quad l_B(1,4) = l_1(2) + l_4(1) = 5.$$

Similarly, Table II depicts the latencies for every possible state of the game. It is clear that the point $(\{1,2\},\{1,4\})$ is the unique NE that this game admits.

TABLE II: Latency computation for each state of the game.

Player A \ Player B	{1,4}	{2,3}
{1,2}	(4,5)	(6,8)
{3,4}	(9,7)	(8,7)

II. Formal Definition of Congestion Games and Extensions

In this section, we will formally define congestion games and we will present two well known extensions of them.

The following is a formal definition of a congestion game, as introduced by Rosenthal [1] and it is usually referred to as *standard congestion game* (or general congestion game).

Definition 1: A standard congestion game G is a tuple (N, R, S_i, l_r), where

$N = \{1, \ldots, n\}$ is a set of n players,

$R = r_1, \ldots, r_m$ a set of m resources,

S_i: the strategy space of player i, and

l_r: a non-decreasing latency function associated with resource r.

A state of the game G is a vector $S = (s_1, \ldots, s_n)$ where player i chooses strategy $s_i \in S_i$. Let $x_r(S)$ express the number of players sharing resource r in state S. Then, the latency $l_i(S)$ of player i is the sum of the latencies of the resources the player uses, *i.e.*,

$$l_i(S) = \sum_{r \in s_i} l_r(x_r(S)).$$

Though the notion of standard congestion games is a general model capturing a variety of different scenarios in which players allocate resources, there are real world scenarios that are not taken into account. For that reason, refinements of the original model have been proposed. The most challenging ones are *weighted congestion games* and *player-specific congestion games* that have originally proposed by Milchtaich in [3].

In weighted congestion games, we additionally associate a weight with each player and define the congestion of a resource as the sum of the players' weights allocating it. Consider the example in the previous section. Actually, the traffic congestion does not depend only on the number of vehicles that share a segment of a road but also on the type of vehicles (obviously, a bus creates more congestion than a bike). So, the congestion on a segment of a road should be defined as the total weight of the vehicles that use a segment on the road rather than their number only.

Formally, in a weighted congestion game, we additionally associate a positive weight w_i with every player i. In this case, we adopt the definition of the congestion of a resource in a particular state of the game in the following way: The congestion $x_r(S)$ on resource r in state S is the sum of the weights of all players sharing resource r in state S, *i.e.*,

$$x_r(S) = \sum_{r \in s_i} w_i.$$

It is worth mentioning that if $w_i = 1$ for each player i, the scenario would be a congestion game as described above.

In player-specific congestion games, the latencies are computed with respect to each player' private latency functions instead of common latency functions applied by all players. For example, different types of vehicles have to conform to different speed limits. Consequently, vehicle-specific speed limits have to be taken into account when computing the travel time along a segment of a road. Formally, for every player i and every resource r, there is a non-decreasing, player-specific latency function $l_r^i(S)$. Then, the latency $l_i(S)$ of player i in state S is given by:

$$l_i(S) = \sum_{r \in s_i} l_r^i(x_r(S)).$$

Some further definitions follow: A congestion game is called *symmetric* if the players' strategy spaces are equal. Otherwise, we call it *asymmetric*. A congestion game is called *singleton congestion game* if all strategies are singleton sets, *i.e.*, each player can use at most one resource. A congestion game is called *network congestion game* if the set of resources R is the set of edges in a(n) (un)directed graph $G = (V, E)$ and each player's strategy space is the set of paths that connects a source $a \in V$ with a sink $b \in V$ of the underlying network.

III. State-of-the-Art

In this section, we summarize results regarding the existence of Nash Equilibria in standard, weighted and player-specific congestion games. Given a class of congestion games it is interested to determine efficiently if a particular game possesses a NE. Furthermore, given a class of games which are guaranteed to possess Nash Equilibria, it is challenging to compute a NE efficiently.

Nash Equilibria are the predominant solution concept to congestion games. A frequently applied technique to prove the existence of Nash Equilibria in congestion games is the potential function method by Monderer and Shapley [4]. The existence of such a function guarantees the existence of a NE. A potential function maps every state of a congestion game to an element from a totally ordered set. From the existence of a potential function, we can also conclude that best response dynamics in which players sequentially play best response are guaranteed to converge to a NE.

Standard Congestion Games: Rosenthal [Ros73] proves that every standard congestion game possesses a NE. As far as the convergence of sequential best response dynamics, they are guaranteed to converge. Fabrikant et al. [5] observe that the convergence time of best response dynamics in standard congestion games can be exponential. They show that this negative result still holds in the case of standard symmetric network congestion games. On the positive side, best response dynamics converge quickly in standard singleton congestion games [6]. Additionally, best response dynamics in standard network congestion games with linearly independent paths are guaranteed to converge quickly [7].

Player-Specific Congestion Games: In general, player-specific congestion games may not possess Nash Equilibria [3], [8]. On the other hand, Milchtaich [3] proves that every player-specific congestion game possesses a NE if the players' strategies are singleton sets, although these games are not potential games.

He also observes that best response dynamics in player-specific singleton congestion games can cycle. However, if the players play their best responses in a certain order, then they quickly find a NE.

Gairing et al. [9] consider a class of player-specific singleton congestion games with linear latency functions. They show that in contrast to general player-specific singleton games these games are potential games and thus admit a NE.

Weighted Congestion Games: Similar to player-specific congestion games, weighted congestion games do not necessarily possess Nash Equilibria [10]. Again, on the other hand, weighted singleton congestion games are potential games, and hence every such game admits a NE [11], [5]. Furthermore, Fotakis et al. [10] prove that every weighted network congestion game with linear latency functions possesses a NE, although weighted network congestion games with arbitrary non-decreasing latency functions do not in general. For weighted network congestion games with arbitrary non-decreasing latency functions and for singleton congestion games which are both weighted and player-specific, it is NP-complete to determine if a given game possesses a NE [12].

Even-Dar et al. [11] consider best response dynamics in weighted singleton congestion games under different assumptions on the players' weights and latency functions, as well as the way the players are selecting to sequentially play a best response. For most scenarios they prove pseudopolynomial bounds on the convergence time.

IV. Applications of Congestion Games in Radio Resource Management in Wireless Networks

Wireless networks consist of entities that share a portion of the spectrum, e.g., Access Points (APs), Base Stations (BSs), and Mobile Nodes (MNs). These are competitors in the sense that their Quality of Service (QoS) targets may not be simultaneously achievable. This is an important problem that arises quite often, even in small topologies where few entities coexist [13]. A game theoretic approach seems suitable to model the interaction among these directly interfering entities and has already successfully been applied in various cases [13], [14]. Even though game theory has been used to model power control in wireless networks, there is still much unexplored space and a lot of work to be done, particularly considering modern environments with independent agents, such as Wi-Fi APs operating in unlicensed spectrum, cognitive radio environments, and ubiquitous wireless sensor networks.

Radio spectrum is a valuable resource that is used in many aspects of the economy and the technology sector. Though, in principle, non-exhaustible, it is limited since it can be neither stored nor exported from country to country. Consequently, spectrum can become easily congested even in a small wireless communication system [14]. Radio Resource Management is a key functionality to utilize the radio spectrum in an efficient way. This can be achieved by adjusting: transmission power, channel, modulation scheme, and other transmission parameters.

Congestion games seem a prominent candidate to model resource competition in wireless networks. The key assumption underlying the standard congestion game model is that all users have an equal impact on the congestion. This means that all that matters is the total number of users that utilize the same resource. This however is not true in wireless networks. Specifically, if we consider channels as resources, then a user's payoff which will be a function of the channel that the user will choose to use depends on two factors: Who the other users are and how much interference it receives from them. We can clearly assume that if all other simultaneous users are sufficiently far away, then sharing the same channel may not cause any performance degradation.

In [15], [16] and [17], the authors consider a wireless communication system with multiple users, each having access to a common set of channels. A user can only access one channel at a time, but he can switch between channels. If multiple users access the same channel at the same time, they will experience degraded performance due to the interference. This problem has recently received increasing interest from the research community, particularly in the context of cognitive radio networks whereby devices are expected to have flexibility in sensing channel availability and moving to better frequencies. A natural way of studying such a system is to model it as a congestion game.

In [15], the authors introduce a concept called "resource expansion", where they define virtual resources as a combination of a spectral and a spatial component. With this definition, a resource is allowed to be reused among non-interfering users. This is a generalization of the original congestion game definition, as the former reduces to the latter if every user interferes with every other user. They cast that problem as a standard congestion game and they apply properties directly associated with them.

In [16] and [17], the authors generalize the notion of "resource expansion". They present an interference graph describing the congestion relationship among users. A user's payoff is a function of the total number of users who are using the same resource and are within its interference set (*i.e.*, connected to it by edges). Therefore, resources are reusable beyond a user's interference set. The authors have shown that when there are only two resources and users can only use one at a time, then a NE exists. This does not hold when there are three or more resources. Moreover, when all resources correspond to the same latency to users (e.g., all channels are of the same bandwidth and/or data rate for all users), then the game possesses a NE.

In [18], the authors study a cellular network consisting of several BSs and MNs. Each BS operates in a different frequency. A MN is free to switch from BS to BS, but can connect to only one BS at a time. Moreover, all MNs have a common signal to interference ratio (SIR) target. The authors propose a distributed algorithm for the combined base station association and power control problem. They model this as a player-specific singleton congestion game. They prove that a NE exists. However, these Nash Equilibria may be far from optimal in terms of minimizing the sum of power consumptions overall the mobiles. They then study a non-atomic version of the system. In a non-atomic game, the number of players limit to infinite and this has the consequence that an incoming mobile or an outgoing mobile affects the congestion in a negligible way. The reason for this transformation of the game is the existence of various positive results in terms of improving/optimizing the efficiency of a NE in games networks with a continuum of users. They propose a non-atomic congestion game model of the system and they show that, under some assumptions, there is a pricing mechanism that can lead to system optimal associations and power allocations. Inspired by these results, they try to apply them for the atomic congestion game so as to improve the system performance.

In a subsequent submitted version [19], the authors also study a network with collocated MNs and model it as a weighted congestion game, as well as a network with collocated BSs and model it as a player-specific weighted congestion game.

In [13], we discuss an integration of the classic Foschini-Miljanic (FM) power control algorithm [20], with bargaining that aims at minimizing the number of unsatisfied entities in cases where it is impossible for all of them to achieve their targets (simultaneously). Our Bargaining Foschini-Miljanic (BFM) algorithm finds a feasible solution in many of these challenging cases through negotiations in a pair of unsatisfied nodes, where one offers to the other some reward with a view to decreasing its transmission power so that the former could achieve its target. Comparing BFM with the previously developed Trunc(ated) FM approach [21], where the weakest node is forced to power off completely, BFM finds an equal number of solutions, however it is more fair, flexible and more realistic in modern (less rigid) environments. In [22], we revisit our BFM algorithm and extend [13], in the following ways:

- we introduce the Distributed Bargaining Foschini-Miljanic algorithm (DBFM), a distributed version of BFM;
- DBFM also generalizes BFM, permitting negotiations among all entities that have not achieved their targets (in order to decide their new transmission power levels), instead of just two entities, which was the case for BFM;
- we further examine and justify our findings on the long term fairness property of (D)BFM (i.e., the rotation of the links that satisfied their targets) in conjunction with the initial budget allocation of the nodes;
- we observe the equivalence of our model to a weighted congestion game.

V. Further Ideas for Applications of Congestion Games in Radio Resource Management in Wireless Networks

In this section, we write down ideas for extensions of the papers that were discussed in the previous section.

One key advantage of congestion games and their extensions is that they are general enough to be applied in every type of wireless network. For example, to: (i) Wi-Fi networks of different operators in the same area, (ii) Femtocells with APs installed in adjacent apartments, (iii) classical cellular networks, where BSs/MNs transmit and interfere with each other, etc. Moreover,they can lead to distributed approaches, which are appealing for future wireless networks because they allow flexible (self-) management.

Concerning the works [15], [16] and [17], it will be very interesting to cast the game as player-specific congestion game or a weighted congestion game and apply properties directly associated with them instead of adopting custom variations of congestion games. Moreover, a limitation in these models is that each player has a fixed transmit power. It would be nice to expand it for power controlled systems. Power control, i.e., controlling the transmission power (in order to achieve a specific target), is a widely applied dynamic radio resource management technique, which is generally adopted for at least one of the following reasons: (i) to mitigate interference in order to increase the capacity of the network, (ii) to conserve energy in order to prolong battery life and-nowadays-to green the Internet/mobile networks, and (iii) to adapt to channel variations in order to support Quality of-Service (QoS).

Concerning the works in [18] and [19], the assumption that each BS operates in a different frequency band could be revised. Moreover, a different social metric could be used.

Finally, concerning our previous works [13], [22], where we have applied a negotiation-based distributed power control algorithm and noticed its equivalence to a weighted congestion game, a natural extension is to apply directly properties of the weighted congestion games.

VI. Acknowledgment

This research was funded by the Research Centre of the Athens University of Economics and Business.

References

[1] R. Rosenthal, "A class of games possessing pure-strategy nash equilibria," *International Journal of Game Theory*, vol. 2, no. 1, pp. 65–67, 1973.

[2] P. Ambrosio and C. Ventre, "Congestion and crowding games," in *Survey within the GI-Seminar: Game Theoretic Analyses of the Internet*, 2004.

[3] I. Milchtaich, "Congestion games with player-specific payoff functions," *Games and Economic Behavior*, vol. 13, pp. 111–124, 1996.

[4] D. Monderer and L. Shapley, "Potential games," *Games and Economic Behavior*, vol. 14, pp. 124–143, 1996.

[5] A. Fabrikant, C. Papadimitriou, and K. Talwar, "The complexity of pure nash equilibria," in *Proceedings of the Thirty-Sixth Annual ACM Symposium on Theory of Computing*, 2004.

[6] S. Ieong, R. McGrew, E. Nudelman, Y. Shoham, and Q. Sun, "Fast and compact: a simple class of congestion games," in *Proceedings of the 20th National Conference on Artificial intelligence*, 2005.

[7] D. Fotakis, "Congestion games with linearly independent paths: Convergence time and price of anarchy," *Algorithmic Game Theory*, pp. 33–45, 2008.

[8] I. Milchtaich, "The equilibrium existence problem in finite network congestion games," in *Proceedings of the 2nd International Workshop on Internet and Network Economics (WINE)*, 2006.

[9] M. Gairing, B. Monien, and K. Tiemann, "Routing (un-) splittable flow in games with player-specific linear latency functions," in *Proceedings of the 33rd International Conference on Automata, Languages and Programming*, 2006.

[10] D. Fotakis, S. Kontogiannis, and P. Spirakis, "Selfish unsplittable flows," *Theoretical Computer Science*, vol. 348, no. 2, pp. 226–239, 2005.

[11] E. Even-Dar, A. Kesselman, and Y. Mansour, "Convergence time to nash equilibria," in *Proceedings of the 30rd International Conference on Automata, Languages and Programming*, 2003.

[12] J. Dunkel and A. Schulz, "On the complexity of pure-strategy nash equilibria in congestion and local-effect games," in *Proceedings of the 2nd International Workshop on Internet and Network Economics (WINE)*, 2006.

[13] V. G. Douros, G. C. Polyzos, and S. Toumpis, "A bargaining approach to power control in networks of autonomous wireless entities," in *Proceedings of the 8th ACM International Workshop on Mobility Management and Wireless Access (MobiWAC)*, 2010.

[14] Q. Zhao and B. Sadler, "A survey of dynamic spectrum access," *IEEE Signal Processing Magazine*, vol. 24, no. 3, pp. 79–89, 2007.

[15] M. Liu and Y. Wu, "Spectum sharing as congestion games," in *Proceedings of the 46th Annual Allerton Conference on Communication, Control, and Computing*, 2008.

[16] M. Liu, S. Ahmad, and Y. Wu, "Congestion games with resource reuse and applications in spectrum sharing," in *Proceedings of the International Conference on Game Theory for Networks (GameNets)*, 2009.

[17] C. Tekin, M. Liu, R. Southwell, J. Huang, and S. Ahmad, "Atomic congestion games on graphs and their applications in networking," *IEEE/ACM Transactions on Networking, (to appear)*, 2012.

[18] C. Singh, A. Kumar, and R. Sundaresan, "Uplink power control and base station association in multichannel cellular networks," in *Proceedings of the International Conference on Game Theory for Networks, (GameNets)*, 2009.

[19] ——, "Combined base station association and power control in multi-channel cellular networks," *Arxiv preprint arXiv:1112.1892*, 2011.

[20] G. Foschini and Z. Miljanic, "A simple distributed autonomous power control algorithm and its convergence," *IEEE Transactions on Vehicular Technology*, vol. 42, no. 4, pp. 641–646, 1993.

[21] J. Zander, "Distributed cochannel interference control in cellular radio systems," *IEEE Transactions on Vehicular Technology*, vol. 41, no. 3, pp. 305–311, 1992.

[22] V. G. Douros, G. C. Polyzos, and S. Toumpis, "Negotiation-based distributed power control in wireless networks with autonomous nodes," in *Proceedings of the 73rd IEEE Vehicular Technology Conference (VTC-Spring 2011)*, 2011.